

1 (In)determinacy in woody perennials: limits and opportunities
2 for timing growth in a changing climate

3 Frederik Baumgarten¹⁻³, Sally Aitken¹, Yann Vitasse³, Robert D Guy¹, EM Wolkovich¹

4 February 14, 2026

5 ¹ Department of Forest and Conservation Sciences, Faculty of Forestry, University of British Columbia,
6 2424 Main Mall Vancouver, BC Canada V6T 1Z4.

7 ² Department of Environmental Sciences-Botany, University of Basel, Schönbeinstrasse 6, Basel 4056,
8 Switzerland.

9 ³ Swiss Federal Institute for Forest, Snow and Landscape Research WSL, Zürcherstrasse. 111, Bir-
10 mensdorf 8903, Switzerland

11 **Corresponding Author:** Frederik Baumgarten; frederik.baumgarten@unibas.ch;
12 Tel.: +41 61 207 23 37

13 **E-Mail-Addresses:** sally.aitken@ubc.ca; yann.vitasse@wsl.ch; rob.guy@ubc.ca; e.wolkovich@ubc.ca

14 **Type of article:** Perspective

15 **Short running title:** (In)determinate growth strategies of trees

16 **Keywords:** plant growth, tree phenology, shoot extension, indeterminate growers, carbon sequestra-
17 tion, growing season length, drought, genetic programming, phenotypic plasticity, forest productivity

18 **wordnumbers:** abstract: 200; main text: 4005; box1: 404; number of figures: 4; number of tables: 0

19 **number of references:** 82

20 **Authors contribution:** FB and EM initiated this project. FB took a lead in drafting, designed
21 figures and performed phylogenetic analyses. All authors contributed substantially to the writing of
22 the article.

23 **Data accessibility statement:** The data supporting the phylogeny are archived in the following
24 public repository. Link: <https://doi.org/10.5281/zenodo.18637216>.

25 **Conflict of interest statement:** The authors declare no conflict of interest

Abstract

When and how much organisms grow given environmental constraints are fundamental questions in biology, and especially pressing given the need to predict biomass production and carbon sequestration with climate change. While environmental factors such as temperature and water availability directly regulate plant growth, the timing and rate of growth are also governed by a plant's phenological sequence—its genetically programmed developmental stages. This sequence is critical for predicting climate responses but often underappreciated.

Here, we leverage the concept of (in-)determinacy in growth and development—which captures how flexibly plants produce organs from preformed cells versus initiating them *de novo*—to propose a new framework for predicting tree growth responses to climate change.

We propose that: 1) determinate growth is an adaptation to strongly seasonal climates, restricting tissue investment to a short favorable period before dormancy; 2) indeterminate species are more exposed to increasing climatic extremes (spring/fall frost, drought) but may recover more effectively from resulting damage; and 3) species with greater indeterminacy are more likely to benefit from lengthening frost-free growing seasons. Future carbon sequestration may thus depend not only on environmental changes, but also on the degree of determinacy set by intrinsic genetic programming.

Introduction

Investing the right amount of resources in growth at the right time is of critical importance to the survival and fitness of any living organism. The topic of growth strategies and habits has a long history in science, spanning the fields of genetics, genomics, physiology and ecology across the animal and plant kingdoms (Stearns, 1998). At its core lies the concept of determinacy—the classification of organisms as either reaching a fixed size at maturity or continuing to grow throughout their lives. Like mollusks, fish and reptiles, plants add biomass to their primary bodies as long as they live and are therefore considered ‘indeterminate growers’ (Ejzmond *et al.*, 2010).

Various terms have emerged to describe this phenomenon across spatial and temporal scales, e.g. from a single cell to the whole organism, and from a single season to an entire lifetime (McDaniel, 1992; Karkach, 2006). In plants, organs can develop from preformed embryonic tissue (e.g. primordia in overwintering buds), form *de novo*, or both, depending on species and life form. While these underlying developmental and meristematic constraints apply broadly across the plant kingdom, they play a particularly central role in the life cycle of woody perennials, especially trees growing in extra-tropical regions, which must balance growth, survival, and reproduction over long lifespans in highly seasonal environments. As we use it here, (in)determinacy in woody perennials refers to the ability to:

- a) preform organs in one year as a future investment, which is deployed rapidly in the following spring with no further primary shoot growth during that season (determinate strategy);
- b) maintain a continuous or episodic meristem activity by forming new shoot tissue during the current growing season (indeterminate strategy).

In this perspective we revisit classical concepts of determinacy and growth patterns of the shoot apical meristem in woody perennials, generalize them across meristem types and position them within the context of climate change. Although we focus here on woody perennials, the concept of (in)determinacy is broadly applicable across vascular plants, including herbaceous species; trees are used as a model system in which the ecological and climatic consequences of determinacy are particularly pronounced. To provide context, we briefly introduce the fundamental limitations to plant growth in extra-tropical regions by linking external environmental drivers and internal growth control mechanisms.

Seasonality of temperature, soil moisture and light

The further one travels from the equator towards the poles, the more strongly plants are confined to a shrinking time window of opportunity for growth set primarily by low temperatures. Below c. 5 °C, metabolic activity slows to an extent where growth and development largely cease (Schenker *et al.*, 2014; Rossi *et al.*, 2008; Körner, 2008). More importantly, sub-freezing temperatures can cause severe damage to plant tissue if they occur at vulnerable developmental stages, e.g. during or after leaf unfolding, prior to fruit maturation, or before cold acclimation in fall (Sakai & Larcher, 1987; Baumgarten *et al.*, 2023). While annual plant species accommodate their entire life cycle within this window, perennial (woody) species of extra-tropical regions are forced to partition their growing phase seasonally, with periods of activity alternating with periods of dormancy (Delpierre *et al.*, 2016). This is referred to as intermittent or rhythmic (as opposed to continuous) growth.

High temperatures can directly inhibit growth by exceeding species-specific physiological thresholds (O’Sullivan *et al.*, 2017). In addition, increased evaporative demand and reduced precipitation can lower soil moisture, leading to water stress that slows or stalls growth and, in extreme cases, causes leaf scorching or premature senescence (Hsiao, 1973; Pugnaire *et al.*, 1999; Etzold *et al.*, 2021; Estiarte & Peñuelas, 2015). Together, these temperature and soil moisture limitations act as environmental filters, narrowing the period during which growth is possible (Figure 1). Notably, comparable (or even stronger) seasonal constraints occur in seasonally dry tropical forests, where temperatures may remain suitable for growth year-round but multi-month dry seasons impose pronounced water limitation. Across species, leaf flush and renewed growth commonly track the onset of rains and access to plant-available water (including deep soil moisture), making soil water availability a primary driver of seasonal activity in these systems (Borchert, 1994; Ramos *et al.*, 2023; Alvarado *et al.*, 2025).

Light also influences plant growth through two key processes. First, light intensity provides the primary energy source that drives photosynthesis and, consequently, source activity. Second, light carries a wealth of information through its quality and the length of daily light and dark periods (photoperiod), which mediate many physiological processes, including bud set and dormancy transitions (Wang *et al.*, 2024). Both factors may contribute to the common observation that tree growth often peaks near the summer solstice in extra-tropical regions (Rossi *et al.*, 2006; Etzold *et al.*, 2021; Luo *et al.*, 2018) or that a window of temperature sensitivity ‘opens’ around the summer solstice to synchronize reproductive

105 efforts over very large spatial scales (Journé *et al.*, 2024).

106 **Internal programming of plants**

107 Given our relatively robust understanding of environmental influences on plant growth, it may seem
108 straightforward to predict how plants will grow under current and future climatic conditions. Yet,
109 this has proven challenging, particularly when predicting responses under scenarios with extended, cli-
110 matically favorable growing seasons (Zohner *et al.*, 2021). Here we propose that accurate predictions
111 require incorporating another key, but often overlooked, factor: internal growth control—the geneti-
112 cally encoded developmental program implemented through hormonal and gene-regulatory networks,
113 that may prevent further growth *despite* favorable environmental conditions.

114
115 While plants have evolved numerous morphological adaptations to tolerate or avoid harmful environ-
116 mental conditions, most temperate and boreal woody perennials cope with seasonal fluctuations in
117 temperature and moisture by temporally escaping these conditions. They do this by aligning their
118 life-history events (phenology) with a cycle of growth and dormancy that balances survival, reproduc-
119 tion, and growth over the long lifespan typical of many tree species (Delpierre *et al.*, 2016). Hence,
120 the phenological sequence can impose abrupt internal switches in resource allocation—from vegetative
121 growth to reproduction (flowering, fruit maturation) and storage (Stearns, 1989; Chapin *et al.*, 1990).
122 These transitions act as additional internal filters, further narrowing the effective window during which
123 vegetative growth can occur, regardless of environmental potential (Figure 1).

125 **The concept of (in)determinate growth**

126 In annual plants, growth ends with the production of flowers to form fruits and seeds: a signal in the
127 apical meristem triggers a sudden switch in resource investment from vegetative growth to building
128 a reproductive structure with no point of return, signaling the end of the life-cycle (Poethig, 2003;
129 Huijser & Schmid, 2011). A shoot axis that terminates in a flower or other modified structure and
130 ceases meristematic activity is considered ‘determinate’ (Barthélémy & Caraglio, 2007). In woody
131 perennial species, however, flowers are typically produced on lateral shoots or buds, preserving the
132 main vegetative axis and allowing continued structural expansion. However, ‘determinate growth’ also
133 refers to a temporal suspension of the meristem, which leads to a time lag between the formation of

134 organs and their expansion (Kozłowski, 2012; Hallé *et al.*, 1978).

135

136 In their first year, all seedlings exhibit indeterminate growth. In subsequent years, many temperate
137 and boreal tree species preform their new shoot increments during the previous growing season, over-
138 wintering them as primordial shoots with leaves and internodes in hardened buds to be ‘ready-to-go’
139 when spring arrives. Once the leaves flush and the preformed internodes elongate, species differ in
140 how long the shoot apical meristem remains competent to continue organ production (Figure 2). Some
141 species exhibit strongly determinate shoot extension, in which the terminal meristem is intrinsically
142 downregulated soon after the spring flush and remains inactive for the rest of the season, consistent
143 with a prolonged internal suppression that is hormonally mediated (paradormancy; Lang *et al.*, 1987;
144 Figure 4; e.g. *Fraxinus* spp., *Acer* spp. and *Prunus* spp.). At the other extreme, highly indetermi-
145 nate species maintain an active shoot apical meristem and can continue producing new leaves and
146 internodes on top of the preformed shoot as long as conditions remain favorable, sometimes stretching
147 their indeterminate growth well into autumn until low temperature eventually forces them to stop
148 (e.g. *Sequoia* spp., *Cupressus* spp. and *Alnus* spp.). Between these extremes, many species occupy an
149 intermediate position: they continue to produce neoformed leaves and internodes after the preformed
150 increment, but growth cessation is often coordinated by seasonal cues—frequently photoperiod acting
151 through hormonal regulation (e.g. *Pinus* spp., *Betula* spp. and *Populus* spp.; Maurya & Bhalerao,
152 2017; Triozzi *et al.*, 2018; Böhlenius *et al.*, 2006). Indeterminate growth can occur either through
153 continuous activity of the shoot apical meristem forming new leaves and internodes without buds (free
154 growth), or through the formation and subsequent flushing of buds mid-season without a period of
155 dormancy (second flushing or lammas growth; Figure 2).

156

157 Although this concept is often presented dichotomously (Kozłowski & Pallardy, 1997; Lechowicz, 1984),
158 but see Kikuzawa, 1983; Damascos *et al.*, 2005), with species classified as either determinate or inde-
159 terminate growers, most species likely exist along a gradient with numerous intermediate forms. For
160 instance, many temperate oaks (*Quercus* spp.) are considered determinate growers, but can exhibit
161 multiple flushes. Similarly, Douglas-fir (*Pseudotsuga menziesii*) gradually adopts a more determinate
162 growth habit with maturity (Borchert, 1976; Heuret *et al.*, 2006). Importantly, the degree of determi-
163 nacy can be quantified at the species level, at least for the shoot apical meristem (Box 1). Figure 4
164 provides a literature-based classification of determinacy for some common species, many of which fall

into an intermediate position.

Evolutionary trade-offs: the cost of certainty and flexibility

In most temperate and boreal ecosystems, determinate and indeterminate growth strategies co-occur, indicating that neither strategy is universally superior. Instead, they reflect contrasting solutions to a fundamental trade-off between growth certainty and growth flexibility under seasonal and variable environments.

Indeterminate species are characterized by continued or episodic activity of the shoot apical meristem during the growing season. This does not necessarily imply faster intrinsic growth rates, but rather a greater capacity to extend growth in time when conditions remain favorable. Such temporal flexibility allows individuals to exploit transient resource availability and recover from partial damage, traits that are advantageous in disturbed or early-successional environments. Accordingly, indeterminate growth is frequently associated with early successional species (Marks, 1975). However, maintaining active meristems later into the season also increases exposure to environmental stressors such as summer drought and early autumn frost, thereby raising the risk of tissue loss or incomplete maturation (Figure 4). As a result, indeterminate growth strategies often involve higher interannual variability in performance and, in some cases, shorter tissue or organismal lifespans (Millet *et al.*, 1999; Brien *et al.*, 2020).

In contrast, determinate species preform a substantial portion of their shoot growth in advance and deploy this tissue during a restricted period of the growing season. This strategy limits short-term plasticity, because the year's shoot-growth potential is set during bud formation in the preceding year and is largely realized by expanding preformed organs (Gordon *et al.*, 2006). At the same time, it reduces exposure of sensitive tissues to late-season stress and synchronizes growth with the most predictable and favorable part of the season. As a consequence, determinate growth strategies are often associated with stronger local adaptation and reduced sensitivity to short-term climatic fluctuations (Leites & Benito Garzón, 2023).

These contrasting strategies help explain the coexistence of determinate and indeterminate species

195 within the same landscapes. For example, western larch (*Larix occidentalis*) and Douglas-fir (*Pseudotsuga menziesii*) frequently co-occur despite marked differences in growth determinacy. Western larch exhibits more flexible, indeterminate shoot growth and performs well in environments shaped by frequent disturbance, whereas Douglas-fir shows more determinate growth and stronger local adaptation, conferring stability under predictable seasonal regimes (Roskill & Aitken, 2024). Such differences suggest niche differentiation along gradients of disturbance frequency, environmental stress, and climatic predictability, rather than outcomes driven by straightforward competitive dominance.

202

203 Together, these patterns indicate that growth determinacy reflects a trade-off between flexibility and reliability in seasonal environments. Under ongoing climate change—marked by longer growing seasons, greater climatic variability, and increasing environmental stress in many regions—it remains uncertain whether determinate or indeterminate growth strategies will be favored. Resolving both how trees regulate their degree of (in)determinacy and which levels along this trait continuum are advantageous under future conditions will be critical for predicting shifts in species performance, coexistence, and forest composition.

210 The performance of (in)determinacy with climate change

211 Spring warming has advanced the onset of leaf emergence by up to a month compared to pre-industrial times (Vitasse *et al.*, 2022), reflecting an earlier reactivation of shoot apical and cambial meristems under warmer conditions. In contrast, the cessation of growth in autumn has shifted far less than expected given the extension of climatically favorable conditions (Zani *et al.*, 2020; Zohner *et al.*, 2023). As a result, phenological sequences are increasingly observed to shift as a whole toward spring rather than being symmetrically extended at both ends of the growing season (Keenan & Richardson, 2015; Meng *et al.*, 2024). This asymmetric response highlights a key constraint relevant to growth (in)determinacy: while growth onset is highly plastic and environmentally responsive, the duration of meristematic activity later in the season appears more tightly regulated by plant internal processes and differs markedly among species. Such constraints on growth cessation help explain why longer growing seasons do not necessarily translate into increased growth or biomass production (Zani *et al.*, 2020; Bose *et al.*, 2025; Wolkovich *et al.*, 2025).

223

224 Along the continuum of growth determinacy, we hypothesize that species positioned toward the more

determinate end will largely maintain their fixed seasonal growth schedules under climate warming, with limited capacity to extend growth later into the season. In regions where climate change primarily increases heat and water stress during mid to late summer, such species are expected to show reduced annual biomass production, because their restricted growth window increasingly overlaps with unfavorable conditions (Figure 3).

In contrast, we predict that species toward the indeterminate end of the continuum will be more capable of prolonging meristematic activity into late summer and autumn when conditions become favorable again, potentially resulting in higher annual production under some future climate scenarios (Caffarra & Donnelly, 2011; Richardson *et al.*, 2009; Zohner *et al.*, 2020, Figure 3). This capacity for temporal extension of growth is particularly common among early successional species, which tend to leaf out earlier in spring and senesce later in autumn than late-successional species. Their earlier leaf-out is consistent with generally lower chilling and forcing requirements for bud burst and dormancy release (Heide, 1993; Laube *et al.*, 2014; Hu *et al.*, 2022; Baumgarten *et al.*, 2021; Basler & Körner, 2012), and in many cases weaker photoperiod sensitivity (Basler & Körner, 2012).

Extreme climatic events and increasing environmental stress provide an important context for evaluating these predictions. More determinate growth strategies concentrate growth earlier in the season and may therefore escape unfavorable growing conditions by operating within a relatively narrow spring to early-summer window that often coincides with comparatively favorable temperature and soil moisture conditions. Under historical climatic regimes, this strategy provided ample safety margins against late-season stress. Where climate change introduces novel or intensified drought stress within this window, however, determinate species are likely to experience reduced annual production and limited capacity for recovery within the same growing season once canopy development has been compromised (Silvestro *et al.*, 2023, Figure 3).

Species with more indeterminate growth habits often remain green and physiologically active for longer periods, leafing out early in spring and senescing late in autumn, with deciduous species occasionally retaining functional foliage until the first freezing events (Figure 3). As a consequence, a substantial fraction of their growth period coincides with parts of the growing season during which drought stress is projected to intensify under future climates, particularly in water-limited regions (Figure 3). While

this extended activity increases exposure to environmental extremes, it also creates opportunities for compensatory growth when conditions become favorable again.

In essence, the flexible growth schedule of indeterminately growing species may allow plants to: (1) produce tissue better adapted to harsh environmental conditions as it is formed in the current season (e.g., smaller leaves with higher specific leaf area), and (2) catch up and compensate later in the season by another production boost (Stevens & Lindroth, 2005). Mediterranean ecosystems provide a clear example of this flexibility, where many species suspend growth during summer drought but resume meristematic activity in autumn or winter when conditions allow (Volaire *et al.*, 2023). Evidence that trees can take advantage of such a ‘second growing season’ after summer drought has been found in pines in Mediterranean climates through polycyclic flushing—a form of indeterminate growth (Girard *et al.*, 2011, Figure 3.).

We therefore hypothesize that climate change will increasingly disrupt established phenological cycles, favoring species and genotypes that are more plastic in rearranging their activities by resuming growth and/or restoring reserves (e.g., starch, nutrients) later in the year, thereby recovering from and compensating for stress-induced damage and losses. Such differential capacity to exploit newly available growth opportunities or to recover after stress is expected to intensify competition among co-occurring species by altering the timing and magnitude of resource uptake, potentially leading to the reassembly of forest communities toward species occupying more indeterminate positions along the growth continuum. This shift may be reinforced if climate change increases not only mean stress but also climatic variability (van der Wiel & Bintanja, 2021), because flexible (plastic) growth schedules can capitalize on transient favorable windows and compensate after setbacks (Richter *et al.*, 2012; Chevin *et al.*, 2010; Botero *et al.*, 2015).

(In)determinacy beyond leaf tissue

Although (in)determinacy is most clearly defined *sensu stricto* at the shoot apical meristem, its relevance extends beyond primary shoot growth because shoot extension imposes fundamental structural constraints on whole-plant growth. Primary growth governs crown expansion and annual leaf deployment, thereby shaping total leaf area and canopy photosynthetic capacity—key determinants of biomass production that have often been underappreciated in studies of tree growth (Girard *et al.*,

286 2011; Soolanayakanahally *et al.*, 2015). At the same time, primary shoot growth establishes the vas-
287 cular framework within which secondary growth occurs. The size and extent of a shoot increment
288 determine the length and surface area over which cambial activity can later be expressed.
289 Empirical studies suggest close coordination between primary shoot growth and stem or radial growth
290 within seasons, indicating that variation in shoot extension is tightly linked to variation in wood for-
291 mation (Cuny *et al.*, 2012; Zhang *et al.*, 2019).

292

293 Despite its central role in tree development, primary shoot growth remains far less frequently measured
294 than radial growth. This imbalance is striking given that shoot extension occurs at the decimeter
295 scale, while radial growth is often expressed at the micrometer scale. Though radial growth can
296 be monitored continuously and with relative ease using dendrometers or microcoring (see Box 1 for
297 alternative approaches to quantify primary growth).

298 In the vascular cambium, seasonal patterns of cell division and differentiation are temporally structured
299 but cannot be interpreted in terms of preformation or neoformation. Cambial initial cells divide to
300 produce daughter cells that mature into functional xylem and phloem (Valdovinos-Ayala *et al.*, 2022).
301 The timing and duration of cambial cell division constrain the rate and seasonal extent of secondary
302 growth by limiting the number of cells entering differentiation, while subsequent cell maturation im-
303 poses additional carbon and resource demands that can further restrict wood formation (Lupi *et al.*,
304 2010). At the same time, cambial cell production can cease even when environmental conditions remain
305 favorable and the canopy is still photosynthetically active, pointing to internal regulation of secondary
306 growth (Buttò *et al.*, 2020; Arend *et al.*, 2024).

307

308 Below-ground meristems appear to follow a more opportunistic strategy. Root apical meristems can
309 remain active whenever local conditions permit, with rhizotron studies documenting root growth dur-
310 ing winter under suitable soil temperatures (Lyford & Wilson, 1966), suggesting that roots often lack
311 a clearly defined dormant phase (Radville *et al.*, 2016; Marchand *et al.*, 2025). However, the extent to
312 which below-ground growth dynamics reflect direct environmental control, coordination with above-
313 ground growth, or genetically constrained strategies remains unresolved (Abramoff & Finzi, 2015;
314 Makoto *et al.*, 2020; Silvestro *et al.*, 2025; Campioli *et al.*, 2024).

315

316 Together, these observations highlight that while (in)determinacy is most rigorously defined at the

shoot apical meristem, primary shoot growth exerts a disproportionate influence on whole-plant performance by simultaneously shaping canopy source capacity and the structural context for downstream sink activity. As such, determinacy at the shoot level provides a meaningful organizing principle for understanding growth regulation across plant organs, even where analogous processes operate through different mechanisms.

Future directions

A central challenge emerging from this perspective is to better understand how variation along the continuum of growth (in)determinacy shapes tree performance under changing climatic conditions. Rather than representing discrete strategies, degrees of determinacy define bounded trait spaces that differ among species but can vary with age, provenance, and environmental context. While the degree of determinacy is largely constrained at the species level, evidence from provenance trials shows meaningful intraspecific variation; for example, populations from lower latitudes often exhibit higher frequencies of second flushes than those from northern origins (Rudolph, 1964). Clarifying how these species-specific bounds and population-level adjustments interact with warming, drought, and increasing climatic variability will be critical for understanding how forest communities respond and reassemble under climate change. Extending this framework beyond woody perennials to include herbaceous species and perennial grasses may further help test the generality of (in)determinacy as a developmental constraint on growth timing, while highlighting differences arising from life form and the absence of secondary growth.

One key knowledge gap concerns how different degrees of (in)determinacy help trees avoid or buffer environmental stress while still maintaining the capacity to resume growth, repair tissues, replenish reserves, and compensate for earlier losses within the same season. Resolving the trade-off between escaping unfavorable conditions and exploiting longer or intermittently favorable growing periods is essential for anticipating how forest communities will change over the coming decades.

Understanding how universally tissues within species are formed determinately or indeterminately could provide insight into the evolution of this trait, which could be aided by cross-species analyses. Both growth forms seem to occur across most species of the same family or clade and hence appear to have evolved repeatedly in different groups (Figure 4), making it difficult to speculate on which

347 strategy is ancestral (but see Hariharan *et al.*, 2016) and how rapidly (in)determinacy can evolve. As
348 data on the degree of (in)determinacy across species accumulates (see Box 1), such analyses could
349 potentially provide insights into this and also aid forecasting for unsampled species.

350

351 Finally, future work should aim to link the temporal dynamics of primary (shoot and root) and sec-
352 ondary (cambial) meristems with patterns of reserve storage and reproductive allocation. Integrating
353 shoot increment formation, wood production, non-structural carbohydrate dynamics, and reproduc-
354 tive effort within a common temporal framework will be essential for understanding whole-tree carbon
355 allocation. Correlating annual tree-ring formation with shoot increments—while accounting for the
356 timing of inter-annual phytomer production and distinguishing between preformed and neoformed tis-
357 sues (Figure 2)—offers a promising path toward such integration and toward improving mechanistic
358 predictions of tree growth and forest carbon dynamics under future climates.

359 Acknowledgments

360 This text emerged from a grant of the Swiss National Foundation to F.B. (grant number P500PB_210943)
361 and was supported by CFI and NSERC Discovery funding to EMW. We thank J. Ngo for designing
362 figure 2.

Bibliography

- Abramoff, R.Z. & Finzi, A.C. (2015). Are above- and below-ground phenology in sync? *New Phytologist*, 205, 1054–1061.
- Alvarado, M.V., Terrazas, T. & Rojas-Leal, A. (2025). Cambial activity in five species of the tropical dry forest: Dynamics of seasonal changes in the cambial region. *Trees*, 39, 51.
- Arend, M., Hoch, G. & Kahmen, A. (2024). Stem growth phenology, not canopy greening, constrains deciduous tree growth. *Tree Physiology*, 44, tpad160.
- Barthélémy, D. & Caraglio, Y. (2007). Plant Architecture: A Dynamic, Multilevel and Comprehensive Approach to Plant Form, Structure and Ontogeny. *Annals of Botany*, 99, 375–407.
- Basler, D. & Körner, C. (2012). Photoperiod sensitivity of bud burst in 14 temperate forest tree species. *Agricultural and Forest Meteorology*, 165, 73–81.
- Baumgarten, F., Gessler, A. & Vitasse, Y. (2023). No risk—no fun: Penalty and recovery from spring frost damage in deciduous temperate trees. *Functional Ecology*, 37, 648–663.
- Baumgarten, F., Zohner, C.M., Gessler, A. & Vitasse, Y. (2021). Chilled to be forced: The best dose to wake up buds from winter dormancy. *New Phytologist*, 230, 1366–1377.
- Böhlenius, H., Huang, T., Charbonnel-Campaa, L., Brunner, A.M., Jansson, S., Strauss, S.H. & Nilsson, O. (2006). CO/FT Regulatory Module Controls Timing of Flowering and Seasonal Growth Cessation in Trees. *Science*, 312, 1040–1043.
- Borchert, R. (1976). The concept of juvenility in woody plants. *Acta Horticulturae*, pp. 21–36.
- Borchert, R. (1994). Soil and Stem Water Storage Determine Phenology and Distribution of Tropical Dry Forest Trees. *Ecology*, 75, 1437–1449.
- Bose, A.K., Etzold, S., Meusburger, K., Gessler, A., Baltensweiler, A., Braun, S., Buchmann, N., Camarero, J.J., Haeni, M., Kahmen, A., Peters, R.L., Sterck, F.J., Tresch, S., Walthert, L. & Zweifel, R. (2025). Decreasing Stem Growth in Common European Tree Species Despite Earlier Growth Onset. *Global Change Biology*, 31, e70318.

388 Botero, C.A., Weissing, F.J., Wright, J. & Rubenstein, D.R. (2015). Evolutionary tipping points in
389 the capacity to adapt to environmental change. *Proceedings of the National Academy of Sciences*,
390 112, 184–189.

391 Brienens, R.J.W., Caldwell, L., Duchesne, L., Voelker, S., Barichivich, J., Baliva, M., Ceccantini, G.,
392 Di Filippo, A., Helama, S., Locosselli, G.M., Lopez, L., Piovesan, G., Schöngart, J., Villalba, R.
393 & Gloor, E. (2020). Forest carbon sink neutralized by pervasive growth-lifespan trade-offs. *Nature*
394 *Communications*, 11, 4241.

395 Buttò, V., Shishov, V., Tychkov, I., Popkova, M., He, M., Rossi, S., Deslauriers, A. & Morin, H. (2020).
396 Comparing the Cell Dynamics of Tree-Ring Formation Observed in Microcores and as Predicted by
397 the Vaganov–Shashkin Model. *Frontiers in Plant Science*, 11.

398 Caffarra, A. & Donnelly, A. (2011). The ecological significance of phenology in four different tree
399 species: Effects of light and temperature on bud burst. *International Journal of Biometeorology*, 55,
400 711–721.

401 Campioli, M., Marchand, L.J., Zahnd, C., Zuccarini, P., McCormack, M.L., Landuyt, D., Lorer, E.,
402 Delpierre, N., Gričar, J. & Vitasse, Y. (2024). Environmental Sensitivity and Impact of Climate
403 Change on leaf-, wood- and root Phenology for the Overstory and Understory of Temperate Decid-
404 uous Forests. *Current Forestry Reports*, 11, 1.

405 Chapin, F.S., Schulze, E. & Mooney, H.A. (1990). The Ecology and Economics of Storage in Plants.
406 *Annual Review of Ecology and Systematics*, 21, 423–447.

407 Chevin, L.M., Lande, R. & Mace, G.M. (2010). Adaptation, Plasticity, and Extinction in a Changing
408 Environment: Towards a Predictive Theory. *PLOS Biology*, 8, e1000357.

409 Cuny, H.E., Rathgeber, C.B., Lebourgeois, F., Fortin, M. & Fournier, M. (2012). Life strategies in
410 intra-annual dynamics of wood formation: Example of three conifer species in a temperate forest in
411 north-east France. *Tree Physiology*, 32, 612–625.

412 Damascos, M.A., Prado, C.H.B.A. & Ronquim, C.C. (2005). Bud Composition, Branching Patterns
413 and Leaf Phenology in Cerrado Woody Species. *Annals of Botany*, 96, 1075–1084.

414 Delpierre, N., Vitasse, Y., Chuine, I., Guillemot, J., Bazot, S., Rutishauser, T. & Rathgeber, C.B.K.
415 (2016). Temperate and boreal forest tree phenology: From organ-scale processes to terrestrial ecosys-
416 tem models. *Annals of Forest Science*, 73, 5–25.

417 Ejsmond, M.J., Czarnoński, M., Kapustka, F. & Kozłowski, J. (2010). How to Time Growth and
 418 Reproduction during the Vegetative Season: An Evolutionary Choice for Indeterminate Growers in
 419 Seasonal Environments. *The American Naturalist*, 175, 551–563.

420 Estiarte, M. & Peñuelas, J. (2015). Alteration of the phenology of leaf senescence and fall in winter
 421 deciduous species by climate change: Effects on nutrient proficiency. *Global Change Biology*, 21,
 422 1005–1017.

423 Etzold, S., Sterck, F., Bose, A.K., Braun, S., Buchmann, N., Eugster, W., Gessler, A., Kahmen, A.,
 424 Peters, R.L. & Vitasse, Y. (2021). Number of growth days and not length of the growth period
 425 determines radial stem growth of temperate trees. *Ecology Letters*.

426 Girard, F., Vennetier, M., Ouarmim, S., Caraglio, Y. & Misson, L. (2011). Polycyclism, a fundamental
 427 tree growth process, decline with recent climate change: The example of *Pinus halepensis* Mill. in
 428 Mediterranean France. *Trees*, 25, 311–322.

429 Gordon, D., Damiano, C. & DeJong, T.M. (2006). Preformation in vegetative buds of *Prunus persica*:
 430 Factors influencing number of leaf primordia in overwintering buds. *Tree Physiology*, 26, 537–544.

431 Guédon, Y., Puntieri, J.G., Sabatier, S. & Barthélémy, D. (2006). Relative Extents of Preformation
 432 and Neoformation in Tree Shoots: Analysis by a Deconvolution Method. *Annals of Botany*, 98,
 433 835–844.

434 Hallé, F., Oldeman, R.A.A. & Tomlinson, P.B. (1978). *Tropical Trees and Forests*. Springer, Berlin,
 435 Heidelberg.

436 Hariharan, I.K., Wake, D.B. & Wake, M.H. (2016). Indeterminate growth: Could it represent the
 437 ancestral condition? *Cold Spring Harbor Perspectives in Biology*, 8, a019174.

438 Heide, O.M. (1993). Daylength and thermal time responses of budburst during dormancy release in
 439 some northern deciduous trees. *Physiologia Plantarum*, 88, 531–540.

440 Heuret, P., Meredieu, C., Coudurier, T., Courdier, F. & Barthélémy, D. (2006). Ontogenetic trends
 441 in the morphological features of main stem annual shoots of *Pinus pinaster* (Pinaceae). *American*
 442 *Journal of Botany*, 93, 1577–1587.

- 443 Hover, A., Buissart, F., Caraglio, Y., Heinz, C., Pailler, F., Ramel, M., Vennetier, M., Prévosto, B.
444 & Sabatier, S. (2017). Growth phenology in *Pinus halepensis* Mill.: Apical shoot bud content and
445 shoot elongation. *Annals of Forest Science*, 74, 39.
- 446 Hsiao, T.C. (1973). Plant responses to water stress. *Annual review of plant physiology*, 24, 519–570.
- 447 Hu, Z., Wang, H., Dai, J., Ge, Q. & Lin, S. (2022). Stronger Spring Phenological Advance in Future
448 Warming Scenarios for Temperate Species With a Lower Chilling Sensitivity. *Frontiers in Plant*
449 *Science*, 13.
- 450 Huijser, P. & Schmid, M. (2011). The control of developmental phase transitions in plants. *Develop-*
451 *ment*, 138, 4117–4129.
- 452 Jin, S., Zhang, W., Shao, J., Wan, P., Cheng, S., Cai, S., Yan, G. & Li, A. (2022). Estimation of Larch
453 Growth at the Stem, Crown, and Branch Levels Using Ground-Based LiDAR Point Cloud. *Journal*
454 *of Remote Sensing*, 2022.
- 455 Journé, V., Szymkowiak, J., Foest, J., Hacket-Pain, A., Kelly, D. & Bogdziewicz, M. (2024). Summer
456 solstice orchestrates the subcontinental-scale synchrony of mast seeding. *Nature Plants*, 10, 367–373.
- 457 Karkach, A.S. (2006). Trajectories and models of individual growth. *Demographic Research*, 15, 347–
458 400.
- 459 Keenan, T.F. & Richardson, A.D. (2015). The timing of autumn senescence is affected by the timing
460 of spring phenology: Implications for predictive models. *Global Change Biology*, 21, 2634–2641.
- 461 Kikuzawa, K. (1983). Leaf survival of woody plants in deciduous broad-leaved forests. 1. Tall trees.
462 *Canadian Journal of Botany*, 61, 2133–2139.
- 463 Körner, C. (2008). Winter crop growth at low temperature may hold the answer for alpine treeline
464 formation. *Plant Ecology & Diversity*, 1, 3–11.
- 465 Kovaleski, A.P., Londo, J.P. & Finkelstein, K.D. (2019). X-ray phase contrast imaging of *Vitis* spp.
466 buds shows freezing pattern and correlation between volume and cold hardiness. *Scientific Reports*,
467 9, 14949.
- 468 Kozłowski, T.T. (2012). *Seed Germination, Ontogeny, and Shoot Growth*. Elsevier.
- 469 Kozłowski, T.T. & Pallardy, S.G. (1997). *Growth Control in Woody Plants*. Elsevier.

470 Kuster, T.M., Dobbertin, M., Günthardt-Goerg, M.S., Schaub, M. & Arend, M. (2014). A Phenological
471 Timetable of Oak Growth under Experimental Drought and Air Warming. *PLoS ONE*, 9, e89724.

472 Lang, G.A., Early, J.D., Martin, G.C. & Darnell, R.L. (1987). Endo-, Para-, and Ecodormancy:
473 Physiological Terminology and Classification for Dormancy Research. *HortScience*, 22, 371–377.

474 Laube, J., Sparks, T.H., Estrella, N., Høfler, J., Ankerst, D.P. & Menzel, A. (2014). Chilling outweighs
475 photoperiod in preventing precocious spring development. *Global Change Biology*, 20, 170–82.

476 Lechowicz, M.J. (1984). Why Do Temperate Deciduous Trees Leaf Out at Different Times? Adaptation
477 and Ecology of Forest Communities. *The American Naturalist*.

478 Leites, L. & Benito Garzón, M. (2023). Forest tree species adaptation to climate across biomes:
479 Building on the legacy of ecological genetics to anticipate responses to climate change. *Global*
480 *Change Biology*, 29, 4711–4730.

481 Luo, T., Liu, X., Zhang, L., Li, X., Pan, Y. & Wright, I.J. (2018). Summer solstice marks a seasonal
482 shift in temperature sensitivity of stem growth and nitrogen-use efficiency in cold-limited forests.
483 *Agricultural and Forest Meteorology*, 248, 469–478.

484 Lupi, C., Morin, H., Deslauriers, A. & Rossi, S. (2010). Xylem phenology and wood production:
485 Resolving the chicken-or-egg dilemma. *Plant, Cell & Environment*, 33, 1721–1730.

486 Lyford, W. & Wilson, B. (1966). Controlled growth of forest tree roots: Technique and application.
487 *Harvard Forest Paper*.

488 Makoto, K., Wilson, S.D., Sato, T., Blume-Werry, G. & Cornelissen, J.H.C. (2020). Synchronous and
489 asynchronous root and shoot phenology in temperate woody seedlings. *Oikos*, 129, 643–650.

490 Marchand, L.J., Gričar, J., Zuccarini, P., Dox, I., Mariën, B., Verlinden, M., Heinecke, T., Prislan,
491 P., Marie, G. & Lange, H. (2025). No winter halt in below-ground wood growth of four angiosperm
492 deciduous tree species. *Nature Ecology & Evolution*, pp. 1–9.

493 Marks, P.L. (1975). On the Relation between Extension Growth and Successional Status of Deciduous
494 Trees of the Northeastern United States. *Bulletin of the Torrey Botanical Club*, 102, 172–177.

495 Maurya, J.P. & Bhalerao, R.P. (2017). Photoperiod- and temperature-mediated control of growth
496 cessation and dormancy in trees: A molecular perspective. *Annals of Botany*, 120, 351–360.

McDaniel, C.N. (1992). Induction and Determination: Developmental Concepts. *Flowering Newsletter*, pp. 3–6.

Meng, F., Felton, A.J., Mao, J., Cong, N., Smith, W.K., Körner, C., Hu, Z., Hong, S., Knott, J., Yan, Y., Guo, B., Deng, Y., Leisz, S., Dorji, T., Wang, S. & Chen, A. (2024). Consistent time allocation fraction to vegetation green-up versus senescence across northern ecosystems despite recent climate change. *Science Advances*, 10, eadn2487.

Millet, J., Bouchard, A. & Édelin, C. (1999). Relationship between architecture and successional status of trees in the temperate deciduous forest. *Écoscience*, 6, 187–203.

Moore, E. (1909). The Study of Winter Buds with Reference to Their Growth and Leaf Content. *Bulletin of the Torrey Botanical Club*, 36, 117–145.

O’Sullivan, O.S., Heskell, M.A., Reich, P.B., Tjoelker, M.G., Weerasinghe, L.K., Penillard, A., Zhu, L., Egerton, J.J.G., Bloomfield, K.J., Creek, D., Bahar, N.H.A., Griffin, K.L., Hurry, V., Meir, P., Turnbull, M.H. & Atkin, O.K. (2017). Thermal limits of leaf metabolism across biomes. *Global Change Biology*, 23, 209–223.

Poethig, R.S. (2003). Phase Change and the Regulation of Developmental Timing in Plants. *Science*, 301, 334–336.

Pugnaire, F.I., Serrano, LUIS. & Pardos, JOSE. (1999). Constraints by water stress on plant growth. *Handbook of plant and crop stress*, 2, 271–283.

Radville, L., McCormack, M.L., Post, E. & Eissenstat, D.M. (2016). Root phenology in a changing climate. *Journal of Experimental Botany*, 67, 3617–3628.

Ramos, D.M., Andrade, J.M., Alberton, B.C., Moura, M.S.B., Domingues, T.F., Neves, N., Lima, J.R.S., Souza, R., Souza, E., Silva, J.R., Espírito-Santo, M.M., Morellato, L.P.C. & Cunha, J. (2023). Multiscale phenology of seasonally dry tropical forests in an aridity gradient. *Frontiers in Environmental Science*, 11.

Richardson, A.D., Hollinger, D.Y., Dail, D.B., Lee, J.T., Munger, J.W. & O’keefe, J. (2009). Influence of spring phenology on seasonal and annual carbon balance in two contrasting New England forests. *Tree Physiology*, 29, 321–331.

524 Richter, S., Kipfer, T., Wohlgemuth, T., Calderón Guerrero, C., Ghazoul, J. & Moser, B. (2012). Phe-
525 notypic plasticity facilitates resistance to climate change in a highly variable environment. *Oecologia*,
526 169, 269–279.

527 Roskilly, B. & Aitken, S. (2024). Weak Local Adaptation to Climate in Seedlings of a Deciduous
528 Conifer Suggests Limited Benefits and Risks of Assisted Gene Flow. *Evolutionary Applications*, 17,
529 e70001.

530 Rossi, S., Deslauriers, A., Anfodillo, T., Morin, H., Saracino, A., Motta, R. & Borghetti, M. (2006).
531 Conifers in cold environments synchronize maximum growth rate of tree-ring formation with day
532 length. *New Phytologist*, 170, 301–310.

533 Rossi, S., Deslauriers, A., Gricar, J., Seo, J.W., Rathgeber, C.B.K., Anfodillo, T., Morin, H., Levanić,
534 T., Oven, P. & Jalkanen, R. (2008). Critical temperatures for xylogenesis in conifers of cold climates.
535 *Global Ecology and Biogeography*, 17, 696–707.

536 Rudolph, T.D. (1964). Lammas Growth and Prolepsis in Jack Pine in the Lake States. *Forest Science*,
537 10, a0001–70.

538 Sakai, A. & Larcher, W. (1987). *Freezing Injuries in Plants*, Springer Berlin Heidelberg, Berlin,
539 Heidelberg, vol. 62, pp. 39–58.

540 Schenker, G., Lenz, A., Körner, C. & Hoch, G. (2014). Physiological minimum temperatures for root
541 growth in seven common European broad-leaved tree species. *Tree Physiology*, 34, 302–313.

542 Silvestro, R., Deslauriers, A., Prislan, P., Rademacher, T., Rezaie, N., Richardson, A.D., Vitasse, Y.
543 & Rossi, S. (2025). From Roots to Leaves: Tree Growth Phenology in Forest Ecosystems. *Current*
544 *Forestry Reports*, 11, 12.

545 Silvestro, R., Zeng, Q., Buttò, V., Sylvain, J.D., Drolet, G., Mencuccini, M., Thiffault, N., Yuan, S.
546 & Rossi, S. (2023). A longer wood growing season does not lead to higher carbon sequestration.
547 *Scientific Reports*, 13, 4059.

548 Smith, S.A. & Brown, J.W. (2018). Constructing a broadly inclusive seed plant phylogeny. *American*
549 *Journal of Botany*, 105, 302–314.

550 Soolanayakanahally, R.Y., Guy, R.D., Street, N.R., Robinson, K.M., Silim, S.N., Albrechtsen, B.R.
551 & Jansson, S. (2015). Comparative physiology of allopatric *Populus* species: Geographic clines in

552 photosynthesis, height growth, and carbon isotope discrimination in common gardens. *Frontiers in*
553 *plant science*, 6, 528.

554 Stearns, S.C. (1989). Trade-Offs in Life-History Evolution. *Functional Ecology*, 3, 259.

555 Stearns, S.C. (1998). *The Evolution Of Life Histories*. Oxford University Press.

556 Stevens, M.T. & Lindroth, R.L. (2005). Induced resistance in the indeterminate growth of aspen
557 (*Populus tremuloides*). *Oecologia*, 145, 297–305.

558 Triozzi, P.M., Ramos-Sánchez, J.M., Hernández-Verdeja, T., Moreno-Cortés, A., Allona, I. & Perales,
559 M. (2018). Photoperiodic Regulation of Shoot Apical Growth in Poplar. *Frontiers in Plant Science*,
560 9, 1030.

561 Valdovinos-Ayala, J., Robles, C., Fickle, J.C., Pérez-de-Lis, G., Pratt, R.B. & Jacobsen, A.L. (2022).
562 Seasonal patterns of increases in stem girth, vessel development, and hydraulic function in deciduous
563 tree species. *Annals of Botany*, 130, 355–365.

564 van der Wiel, K. & Bintanja, R. (2021). Contribution of climatic changes in mean and variability to
565 monthly temperature and precipitation extremes. *Communications Earth & Environment*, 2, 1.

566 Vitasse, Y., Baumgarten, F., Zohner, C.M., Rutishauser, T., Pietragalla, B., Gehrig, R., Dai, J., Wang,
567 H., Aono, Y. & Sparks, T.H. (2022). The great acceleration of plant phenological shifts. *Nature*
568 *Climate Change*, 12, 300–302.

569 Volaire, F., Barkaoui, K., Grémillet, D., Charrier, G., Dangles, O., Lamarque, L.J., Martin-StPaul,
570 N. & Chuine, I. (2023). Is a seasonally reduced growth potential a convergent strategy to survive
571 drought and frost in plants? *Annals of Botany*, 131, 245–254.

572 Wang, Q., Liu, W., Leung, C.C., Tarté, D.A. & Gendron, J.M. (2024). Plants distinguish different
573 photoperiods to independently control seasonal flowering and growth. *Science*, 383, eadg9196.

574 Wolkovich, E.M., Ettinger, A.K., Chin, A.R., Chamberlain, C.J., Baumgarten, F., Pradhan, K., Man-
575 zanedo, R.D. & Hille Ris Lambers, J. (2025). Why longer seasons with climate change may not
576 increase tree growth. *Nature Climate Change*, 15, 1283–1292.

577 Zani, D., Crowther, T.W., Mo, L., Renner, S.S. & Zohner, C.M. (2020). Increased growing-season
578 productivity drives earlier autumn leaf senescence in temperate trees. *Science*, 370, 1066–1071.

- 579 Zhang, Y., Wu, H. & Yang, W. (2019). Forests Growth Monitoring Based on Tree Canopy 3D Recon-
580 struction Using UAV Aerial Photogrammetry. *Forests*, 10, 1052.
- 581 Zohner, C.M., Mirzaghali, L., Renner, S.S., Mo, L., Rebindaine, D., Bucher, R., Palouš, D., Vitasse,
582 Y., Fu, Y.H., Stocker, B.D. & Crowther, T.W. (2023). Effect of climate warming on the timing of
583 autumn leaf senescence reverses after the summer solstice. *Science*, 381, eadf5098.
- 584 Zohner, C.M., Mo, L., Pugh, T.A.M., Bastin, J.F. & Crowther, T.W. (2020). Interactive climate
585 factors restrict future increases in spring productivity of temperate and boreal trees. *Global Change*
586 *Biology*, 26, 4042–4055.
- 587 Zohner, C.M., Renner, S.S., Sebald, V. & Crowther, T.W. (2021). How changes in spring and autumn
588 phenology translate into growth-experimental evidence of asymmetric effects. *Journal of Ecology*,
589 109, 2717–2728.

Box 1: Metrics and proxies related to (in)determinacy

To advance our understanding of growth strategies under climate change, we need not only improved methods for quantifying the degree of (in)determinacy—but also empirical approaches that move beyond a simple binary classification. An organ-level metric at the shoot apical meristem captures the defining features of (in)determinacy and allows it to be explicitly defined and quantified. Temporal dynamics of shoot growth provide a related but indirect proxy that can be applied from individual shoots to crowns and entire forest stands.

Ratio of preformed to neoformed organs: The ratio of the number of leaves (or leaf scars) produced during a growing season to the number of leaf primordia preformed in dormant buds provides a direct, mechanistic, and quantitative measure of shoot-level (in)determinacy. Values close to one indicate growth constrained to preformed organs, whereas values greater than one reflect the production of neoformed organs and thus indeterminate development. Preformation and neoformation have been documented for more than a century (Moore, 1909), and several studies have quantified the number of preformed primordia and compared it with the final number of emerged leaves or phytomers (Damascos *et al.*, 2005; Kikuzawa, 1983; Guédon *et al.*, 2006). Quantification typically relies on bud dissection, but emerging non-destructive approaches such as micro-CT imaging offer promising alternatives for assessing bud structure and preformation (Kovaleski *et al.*, 2019). This metric remains the most conceptually robust approach for defining and quantifying the level of (in)determinacy across species.

Seasonal growth dynamics of the shoot apical meristem: Temporal patterns of shoot elongation provide a complementary perspective on (in)determinacy at the shoot apical meristem. Across species, elongation may occur over a brief period, extend continuously through the growing season, or appear in multiple distinct growth phases, reflecting differences in how apical meristem activity is maintained or reactivated. Such dynamics can be quantified using repeated shoot measurements or phenological observations, and increasingly through high-resolution terrestrial LiDAR and photogrammetric techniques that capture shoot elongation and structural change at the scale of entire crowns or stands (Zhang *et al.*, 2019; Jin *et al.*, 2022). While temporal growth dynamics do not distinguish preformed from neoformed organs directly, they remain closely linked to shoot apical meristem activity and provide a scalable proxy for the seasonal expression of (in)determinacy. Repeated measurements also allow shoot-elongation curves to be reconstructed, whose shape differs systematically among species (single, short sigmoid vs. prolonged or multi-phase extension; Hover *et al.* (2017); Kuster *et al.* (2014)). Specifically, curve-derived parameters such as the duration of shoot expansion relative to the growing season provide a quantitative proxy for the degree of (in)determinacy.

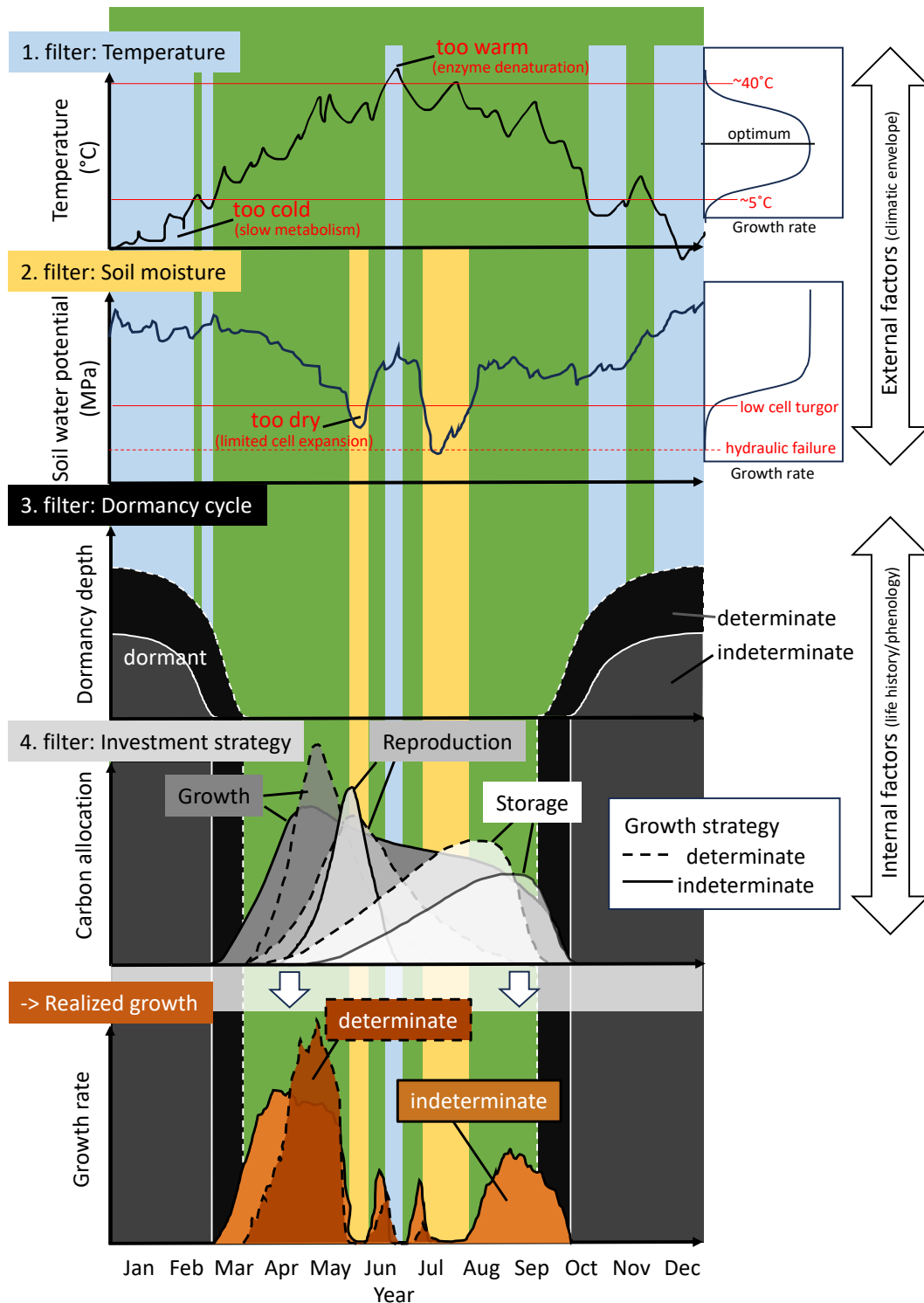


Figure 1: Schematic overview of the possible discrepancy between the potential growing season and the effectively realized vegetative growth (shoot and possibly cambial growth). When environmental factors, such as temperature and soil moisture, exceed growth-promoting thresholds they can act as filters that narrow the window of opportunity available for vegetative growth (potential light and photoperiodic constraints are not shown). The species-specific life history cycle (phenology) can impose another filter by dictating a dormancy cycle and prioritizing developmental processes other than vegetative growth (e.g. flowering, fruit maturation and storage). The degree of (in)determinacy is presented here in two extremes, although we propose to consider this trait as continuous.

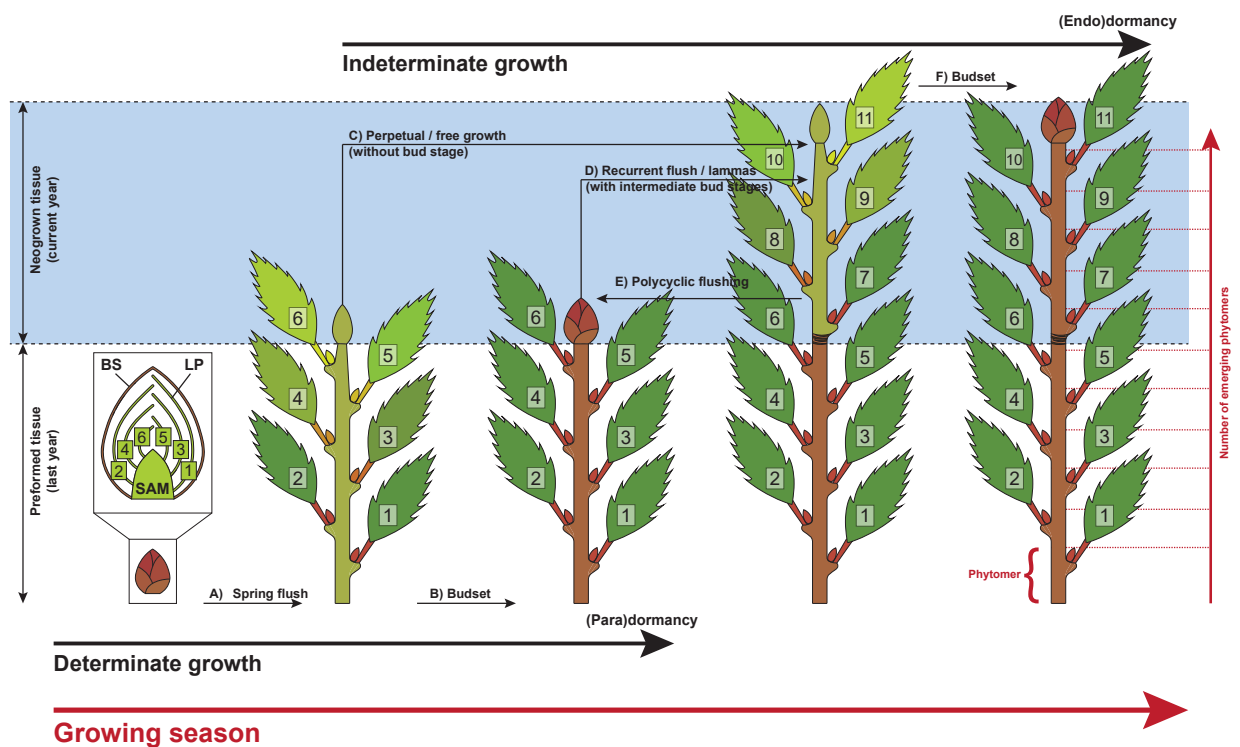


Figure 2: Determinate and indeterminate growth within one growing season for species producing terminal buds. Commonly all tree species deploy buds during their first (spring) flush from prebuilt and overwintering leaf primordia (LP) protected by bud scales (BS; A). Determinate growing species set buds (B) that are under hormonal suppression to inhibit any further activity of the shoot apical meristem – SAM (paradormancy). The basic unit of a shoot is the phytomer which is composed of a node, a leaf, the axillary bud and an internode (see bottom right). Indeterminate growing species continue to produce new phytomers directly (C) or through one (D) to several (E) intermediate bud stage(s). Finally, most species set their terminal buds (F) and enter full dormancy (endodormancy).

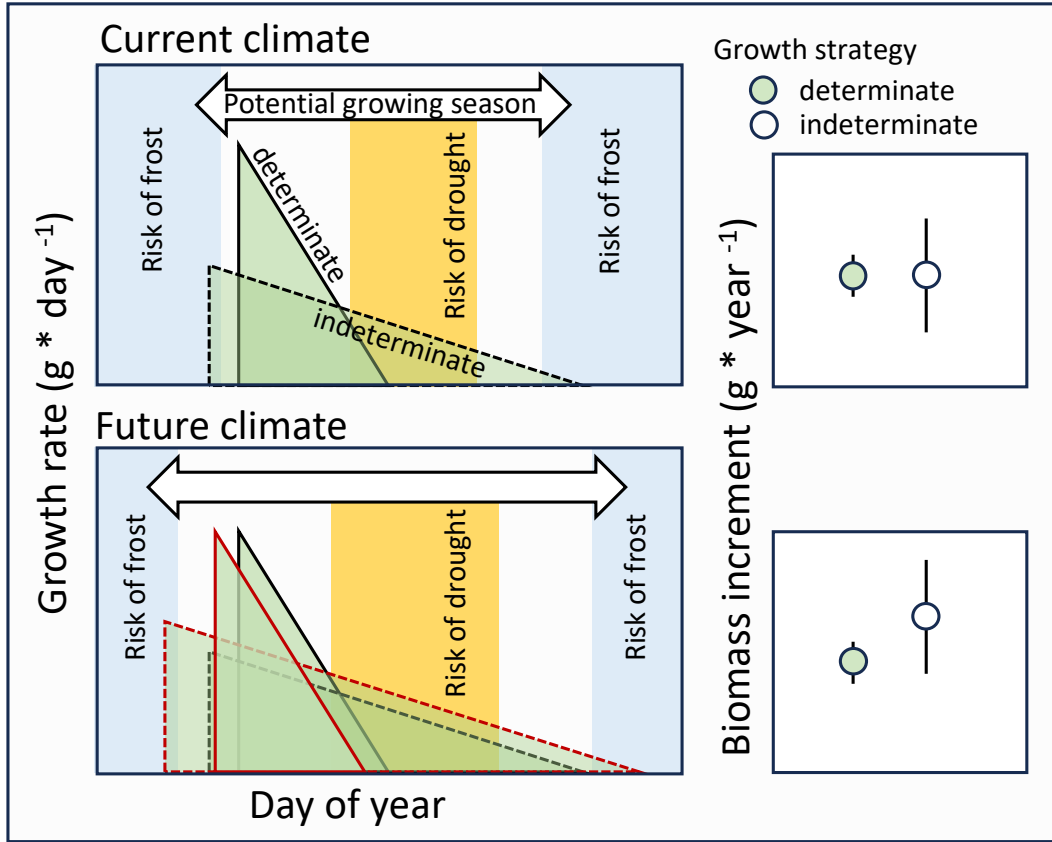


Figure 3: Hypothesized predictions of growth rates under current (upper panels) and future (lower panels) climate for determinate and indeterminate growing species. Note that the indeterminate strategy is more exposed to the risk of frost and drought events while the determinate strategy condenses most growth within a safe period. In the current climate the indeterminate strategy is in balance with benefiting from the full climatic growing season in some years with some drawbacks in other years, resulting in the same mean yearly biomass increment, but with a higher variation (right panel). In a future climate the indeterminate strategy might benefit from longer growing seasons, resulting in an overall higher mean annual biomass increment compared to determinate growers, but this will be dependent on the severity of drought that overlaps the potential growth period of determinate species. Triangle outlines represent the activity window for a current (black) versus a future (red) climate.

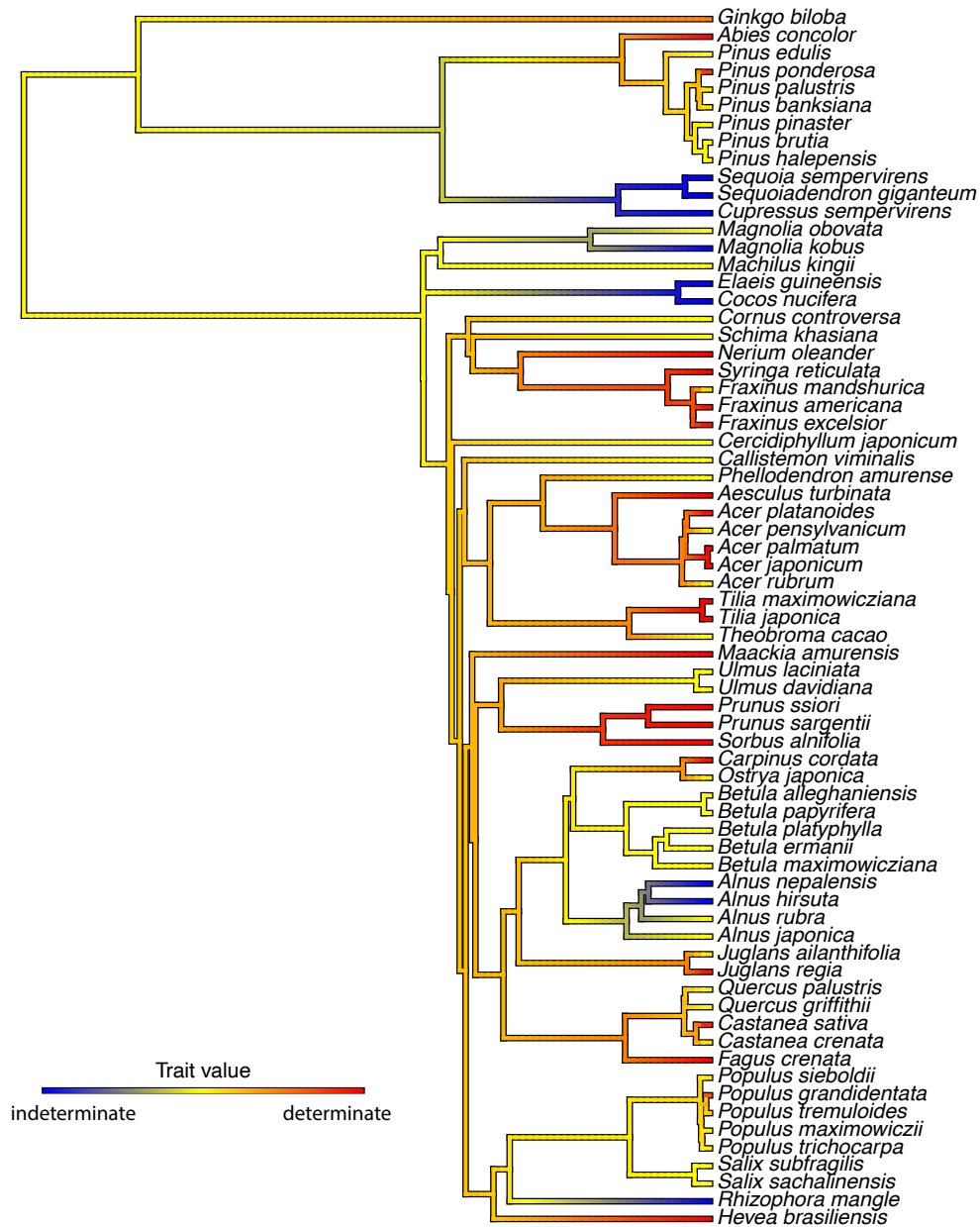


Figure 4: Phylogenetic tree of plant species colored by growth determinacy trait values, visually representing the evolutionary distribution of growth strategies across taxa. This tree includes species for which shoot growth determinacy data were compiled (mainly from Hallé *et al.* (1978) and Kikuzawa (1983)), with the phylogeny pruned from the comprehensive seed plant tree of Smith & Brown (2018) to match the trait dataset. Growth determinacy was scored as indeterminate (blue), intermediate (yellow) and determinate (red), with mean values calculated per species when multiple entries existed. These values were then mapped as a continuous trait onto the phylogeny using the `contMap()` function from the `phytools` R package.